

Exercise5

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```
library(disctr) # A companion package for the book Introduction to Discrete Choice Analysis with `R`
library(dplyr) # A Grammar of Data Manipulation
library(ggplot2) # Create Elegant Data Visualisations Using the Grammar of Graphics
library(htmlwidgets) # HTML Widgets for R
library(kableExtra) # Construct Complex Table with 'kable' and Pipe Syntax
library(mlogit) # Multinomial Logit Models
library(plotly) # Create Interactive Web Graphics via 'plotly.js'
library(stargazer) # Well-Formatted Regression and Summary Statistics Tables
library(tidyr) # Tidy Messy Data

data("mc_commute_wide",
      package = "disctr")

mc_commute_long <- mc_commute_wide %>%
  mlogit.data(shape="wide",
             # Name of column with the choices
             choice = "choice",
             # Numbers of columns with attributes that vary by alternative
             varying = 3:22)

# Function `mFormula()` is used to define multi-part formulas of the form:
# y ~ x | z | w, which in the notation used for the anatomy of utility functions is
# choice ~ alternative vars. with generic coefficients |
#           individual vars. with specific coefficients |
#           alternative vars. with specific coefficients
# In this formula time is one of x variables
f1 <- mFormula(choice ~ time)

# Function `mFormula()` is used to define multi-part formulas of the form:
# y ~ x | z | w, which in the notation used for the anatomy of utility functions is
# choice ~ alternative vars. with generic coefficients |
#           individual vars. with specific coefficients |
#           alternative vars. with specific coefficients
# In this formula `time` is one of x variables and `sidewalk_density` is one of z variables
f2 <- mFormula(choice ~ time | sidewalk_density)

f3 <- mFormula(choice ~ 0 | sidewalk_density | time)

# Function `mlogit()` is used to estimate logit models
# It needs a multi-part formula and a data set in long form
model1 <- mlogit(f1,
                 mc_commute_long)

# Function `summary()` give the summary of data objects,
```

```

# including the output of model estimation algorithms
summary(model1)

##
## Call:
## mlogit(formula = choice ~ time, data = mc_commute_long, method = "nr")
##
## Frequencies of alternatives:choice
##      Car      Cycle      HSR      Walk
## 0.204364 0.034182 0.244364 0.517091
##
## nr method
## 5 iterations, 0h:0m:0s
## g'(-H)^-1g = 0.000102
## successive function values within tolerance limits
##
## Coefficients :
##              Estimate Std. Error z-value Pr(>|z|)
## (Intercept):Cycle  0.366898   0.201434  1.8214  0.06854 .
## (Intercept):HSR    0.707267   0.118104  5.9885 2.118e-09 ***
## (Intercept):Walk   3.212834   0.184169 17.4451 < 2.2e-16 ***
## time              -0.056494   0.005685 -9.9374 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-Likelihood: -768.63
## McFadden R^2:  0.50323
## Likelihood ratio test : chisq = 1557.2 (p.value = < 2.22e-16)
# Note: use chunk option results="asis" to display latex output in pdf
stargazer::stargazer(model1,
                      # Use type = "text", "latex". or "html" depending
                      # on the desired output
                      type = "text",
                      header = FALSE,
                      single.row = TRUE,
                      title = "Estimation results: Model 1")

```

Estimation results: Model 1

Dependent variable:

choice

(Intercept):Cycle	0.367*	(0.201)	(Intercept):HSR	0.707***	(0.118)
(Intercept):Walk	3.213***	(0.184)	time	-0.056***	(0.006)

Observations 1,375 R² 0.503 Log Likelihood -768.633 LR Test
1,557.233*** (df = 4)

=====

Note: $p < 0.1$; $p < 0.05$; $p < 0.01$

“ r # Note: use chunk option results=“asis” to display latex output
in pdf model2 <- mlogit(f2, mc_commute_long)

```
(Intercept):Cycle 0.367* (0.201) (Intercept):HSR 0.707*** (0.118)
(Intercept):Walk 3.213*** (0.184) time -0.056*** (0.006)
```

```
stargazer::stargazer(model2, # Use type = "text", "latex". or "html"
depending # on the desired output type = "text", header = FALSE,
single.row = TRUE, title = "Estimation results: Model 2") ""
Estimation results: Model 2
```

```
=====
Dependent variable: ----- choice
=====
```

```
(Intercept):Cycle -0.306 (0.454)
(Intercept):HSR 0.676*** (0.215)
(Intercept):Walk 2.345*** (0.312)
time -0.056*** (0.006)
sidewalk_density:Cycle 0.027 (0.017)
sidewalk_density:HSR 0.002 (0.009)
sidewalk_density:Walk 0.035*** (0.011)
----- Observations 1,375
```

```
R2 0.509
```

```
Log Likelihood -760.418
```

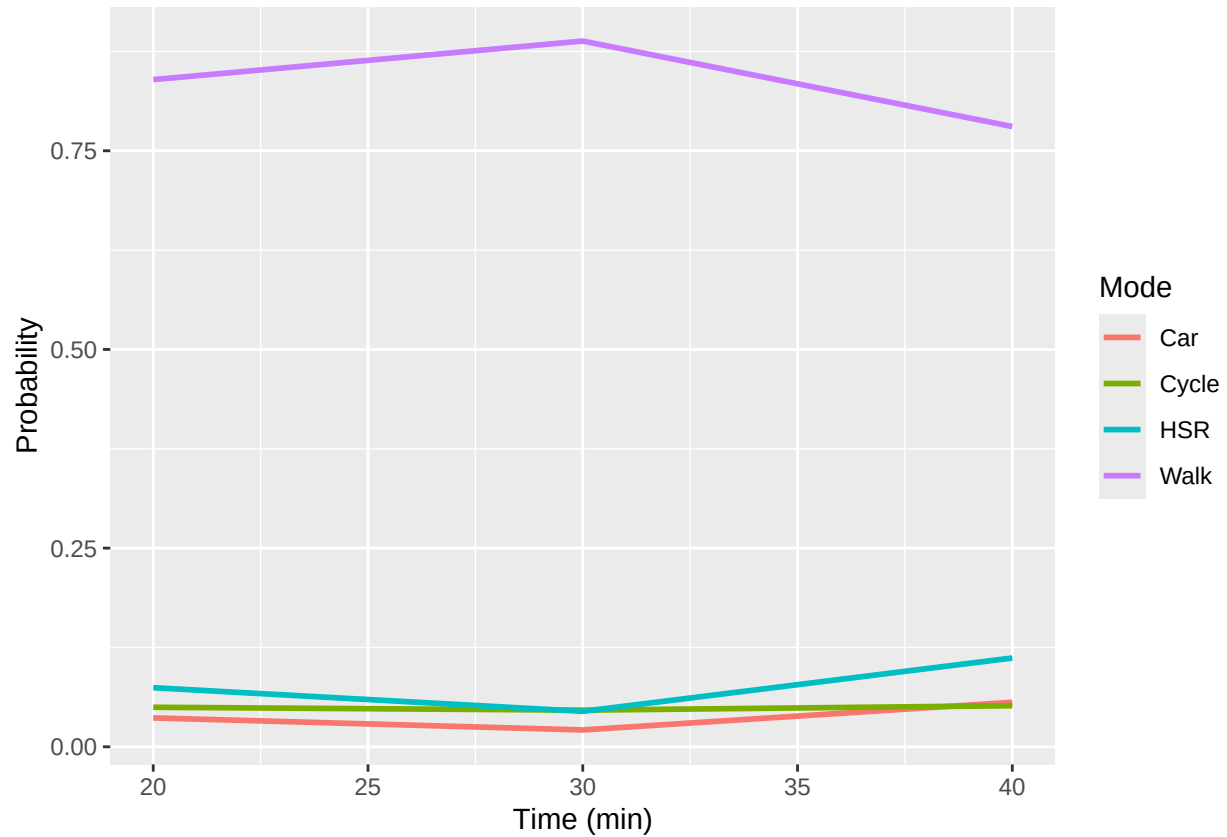
```
LR Test 1,573.662*** (df = 7)
```

```
===== Note: p<0.1;
p<0.05; p<0.01
```

1. In the example in this chapter we estimated the probabilities of choosing different

modes by sidewalk density setting travel time to the in-sample median. Use `model2` to calculate the probability of choosing different modes by in-sample median sidewalk density but now for travel times of 20, 30, and 40 min. Discuss the results.

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



2. Estimate a model using formula f3 (call it model3). Discuss the output of this model.

```
# Note: use chunk option results="asis" to display latex output in pdf
model3 <- mlogit(f3,
  mc_commute_long,
  # Specify the alternative that acts as reference
  reflevel = "Walk")

stargazer::stargazer(model3,
  # Use type = "text", "latex". or "html" depending
  # on the desired output
  type = "text",
  header = FALSE,
  single.row = TRUE,
  title = "Estimation results: Model 3 (alternative: Walk)")
```

Estimation results: Model 3 (alternative: Walk)

Dependent variable:

choice

(Intercept):Car -4.541*** (0.441)
 (Intercept):Cycle -6.971*** (0.853)

```

(Intercept):HSR -3.280*** (0.376)
sidewalk_density:Car -0.027** (0.012)
sidewalk_density:Cycle 0.027 (0.019)
sidewalk_density:HSR -0.026*** (0.009)
time:Walk -0.129*** (0.014)
time:Car 0.044*** (0.013)
time:Cycle 0.231*** (0.059)
time:HSR -0.023*** (0.006)

```

Observations 1,375

R2 0.557

Log Likelihood -684.747

LR Test 1,725.005*** (df = 10)

===== Note: $p < 0.1$;
 $p < 0.05$; $p < 0.01$

3. Use `model3` to calculate the probability of choosing different modes by insample

median sidewalk density but now for travel times of 20, 30, and 40 min. Discuss the results.

```

probs3 <- predict(model3,
  newdata = mc_commute_predict)

```

```

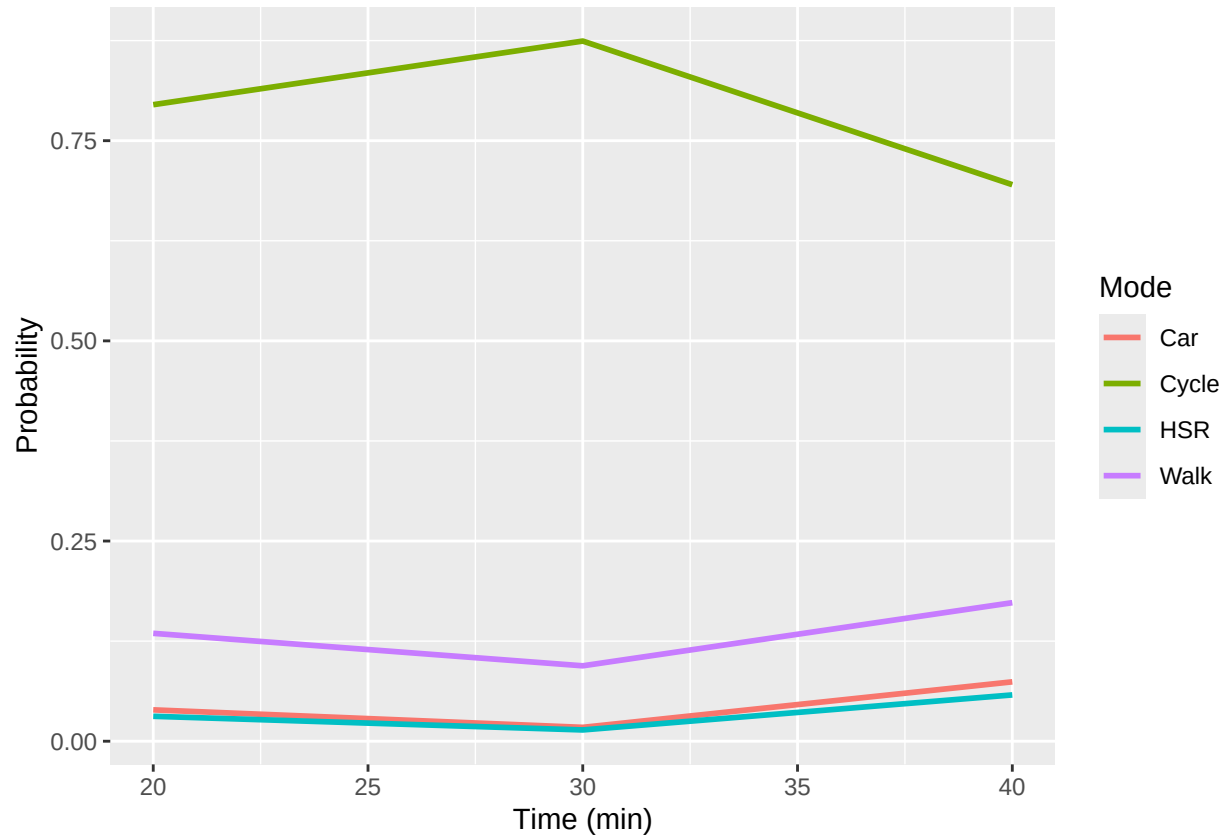
probs3 <- data.frame(time = seq(from = 20,
  to = 40,
  by = 10),
  probs3) %>%
  # Pivot longer all columns _except_ `sidewalk_density`
  pivot_longer(cols = -time,
    # The column names become a new column called "Mode"
    names_to = "Mode",
    # The values are gathered into a single column called
    # "Probability"
    values_to = "Probability")

```

```

probs3 %>%
  # Create ggplot object; map `sidewalk_density` to the y-axis
  # `Probability` to the x-axis, and the color of geometric
  # objects to `Mode`
  ggplot(aes(x = time,
    y = Probability,
    color = Mode)) +
  # Add geometric object of type line with size = 1
  geom_line(size = 1) +
  labs(y="Probability",
    x = expression("Time (min)"))

```



4. In the general case, what is the value of the log-likelihood of the null (Equally Likely) model?

$$LL_{null} = -N * \log J$$

5. Use the likelihood ratio test to compare model3 to model2? Discuss the results.

What restrictions would you need to impose in model3 to obtain model2?

```
lrtest(model2,
       model3)
```

```
## Likelihood ratio test
##
## Model 1: choice ~ time | sidewalk_density
## Model 2: choice ~ 0 | sidewalk_density | time
##   #Df LogLik Df  Chisq Pr(>Chisq)
## 1    7 -760.42
## 2   10 -684.75  3 151.34 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The very small p-value ($< 2.2e-16$) means we strongly reject the null hypothesis, which suggests that Model 2 fits significantly better than Model 1. Since Model 2 has a better fit, this indicates that including “time” as an alternative-specific predictor provides valuable information for modeling “choice”.

Restrictions: setting the alternative var. with specific coefficients $(\beta_2, \beta_3, \beta_4)$ of **model3** to be the same value of β_1 .